ENGINE

6A12

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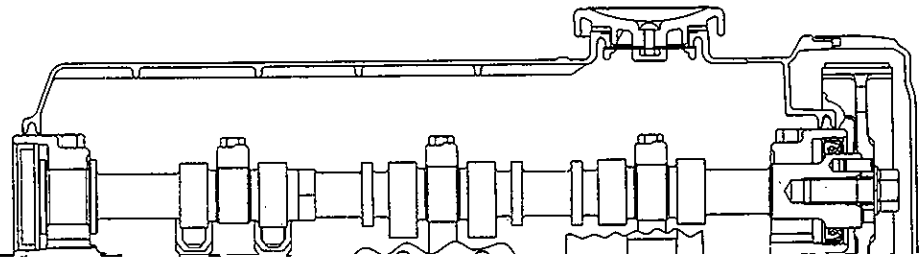
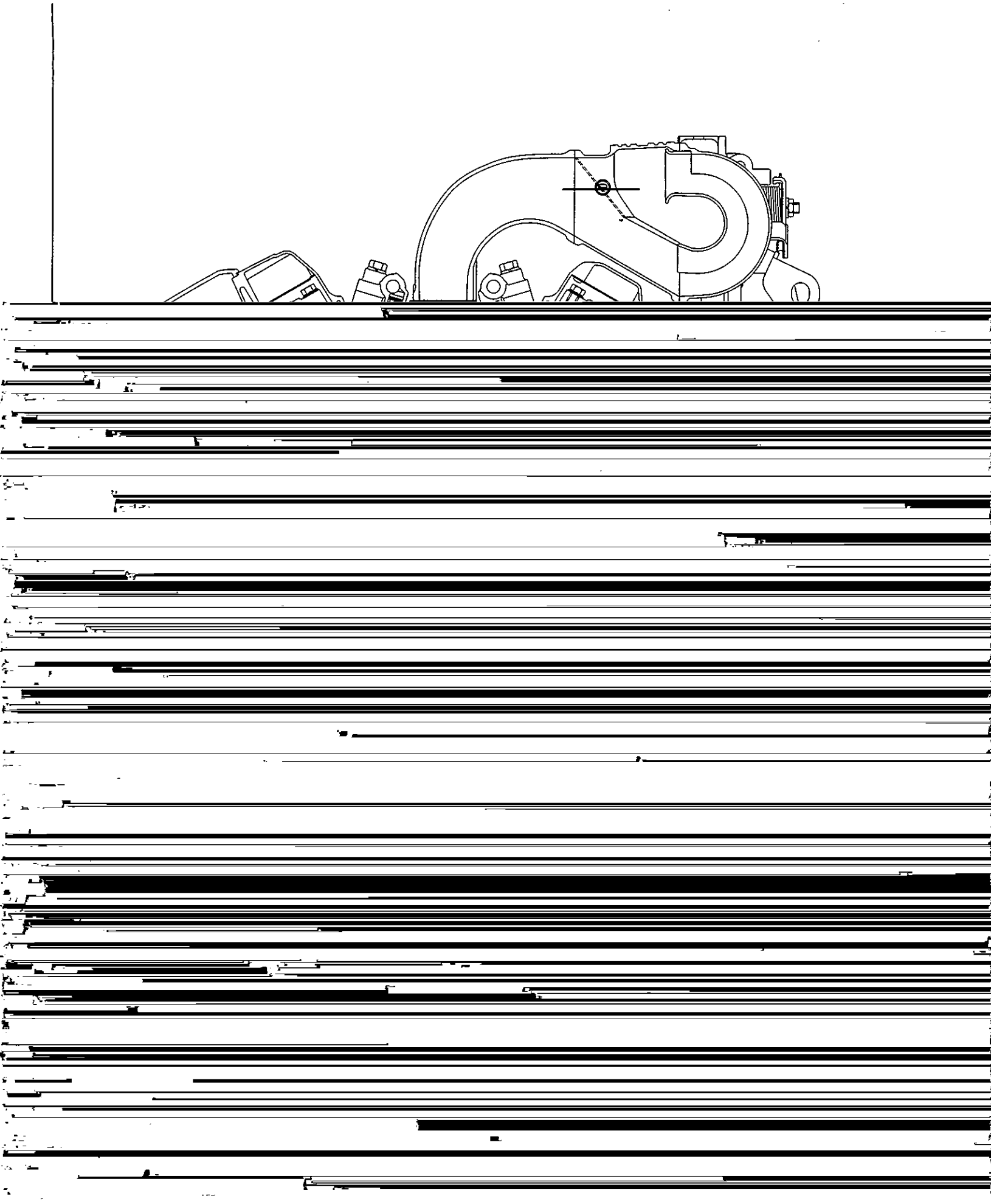
GENERAL SPECIFICATIONS

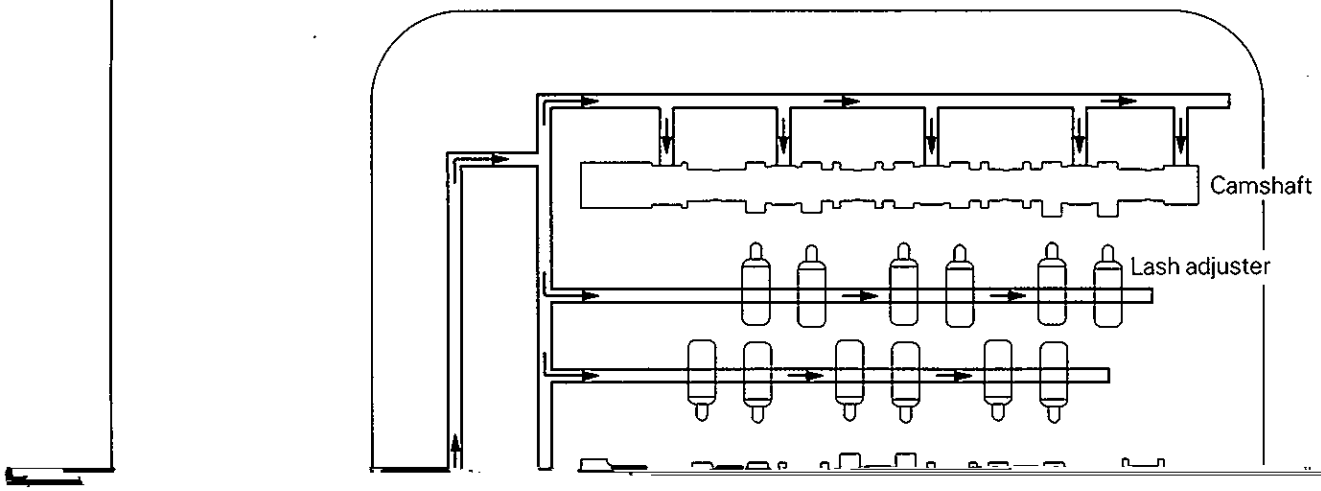
GENERAL INFORMATION

SECTIONAL VIEW





LUBRICATION SYSTEM



NOTES

1. SPECIFICATIONS

GENERAL SPECIFICATIONS

Description	Specifications
Engine model	6A12
No. and arrangement of cylinders	6, V – type
Combustion chamber	Compact type
Total displacement	1,998 cm^3 (121.9 cu.in.)
Cylinder bore x stroke	78.4 x 69 mm (3.09 x 2.72 in.)
Compression ratio	10.0

SERVICE SPECIFICATIONS

mm (in.)

	Standard	Limit
Cylinder head		
Flatness of gasket surface	Less than 0.05 (0.0019)	0.2 (0.008)
Grinding limit		*0.2 (0.008)

mm (in.)

	Standard	Limit
Valve guide		
Overall length		
Intake	45.5 (1.791)	
Exhaust	52.5 (2.067)	
I.D.	6.00 – 6.01 (0.236 – 0.237)	
O.D.	11.06 – 11.07 (0.4354 – 0.4358)	
Service size	0.05 (0.002), 0.25 (0.010) 0.50 (0.020) Oversize	
Valve seat		
Seat angle	44° – 44.5°	

mm (in.)

	Standard	Limit
Flywheel		
Runout		0.13 (0.005)
Oil pump		
Tip clearance	0.06 – 0.18 (0.0024 – 0.0070)	
Body clearance	0.10 – 0.18 (0.0040 – 0.0070)	
Side clearance	0.04 – 0.10 (0.0016 – 0.0039)	
Drive belt		
Deflection		
New belt	8.5 – 10.5 (0.335 – 0.402)	

TORQUE SPECIFICATION

	Torque		
	Nm	kgm	ft.lbs.
Drive belt			
Alternator	45	4.5	33
Alternator bracket	23	2.3	17
Oil level gauge	14	1.4	10
Tensioner bracket	45	4.5	33
Crankshaft bolt	185	18.5	134
Alternator pivot bolt	12	1.2	9
Timing belt			
Timing belt front cover bolt	11	1.1	8

	Torque		
	Nm	kgm	ft.lbs.
Intake and exhaust manifold			

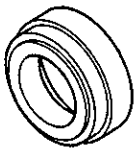
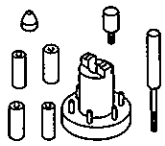
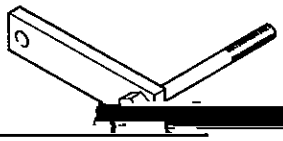
FORM-IN-PLACE GASKET

The engine has several areas where the form-in-place gasket (FIPG) is in use. To ensure that the gasket fully serves its purpose, it is necessary to observe some precautions when applying the gasket. Bead size, continuity and location are of paramount importance. Too thin a bead could cause leaks. Too thick a bead, on the other hand, could be squeezed out of location, causing blocking or narrowing of the fluid feed line. To eliminate the

2. SPECIAL TOOLS

Tool	Number	Name	Use

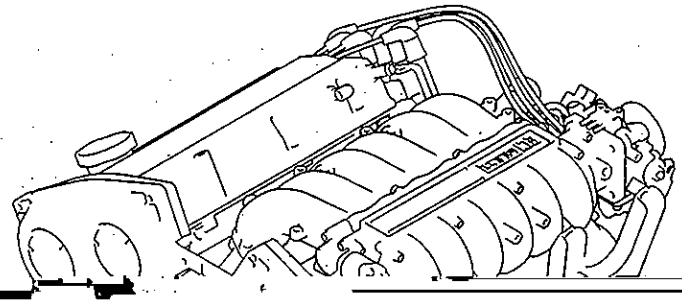
Tool	Number	Name	Use
	MD998717	Crankshaft front oil seal installer	Installation of crankshaft front oil seal
	MD998727	Oil pan remover	Removal of the oil pan

Tool	Number	Name	Use
	MD998777	Camshaft oil seal installer adapter	Installation of camshaft oil seal
	MD998780	Piston pin setting tool	Removal and installation of piston pin
	MD998781	Flywheel stopper	Holding flywheel and drive plate

NOTES

3. DRIVE BELT

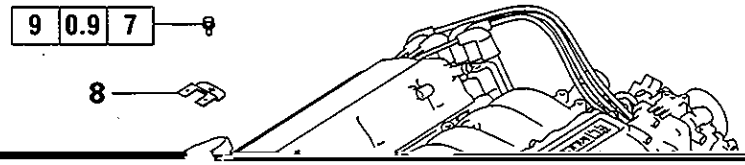
REMOVAL AND INSTALLATION

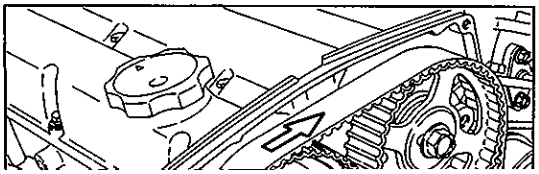


REMOVAL SERVICE POINTS

4. TIMING BELT

REMOVAL AND INSTALLATION

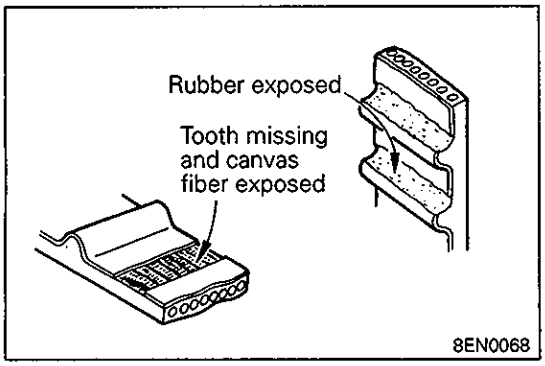




REMOVAL SERVICE POINTS

⟨A⟩ TIMING BELT REMOVAL

(1) Mark the belt running direction for reference in reinstallation.

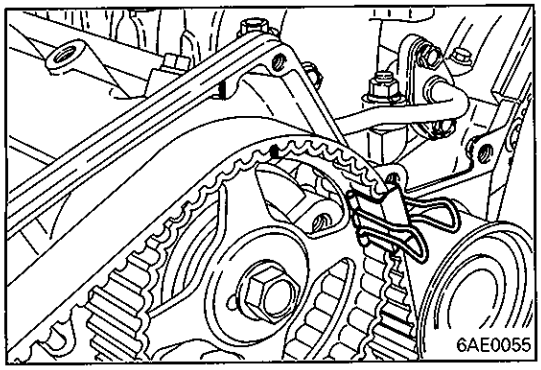


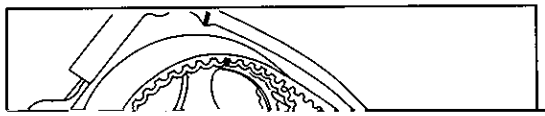
(7) Abnormal wear on teeth.
(8) Missing tooth.

AUTO-TENSIONER

(1) Check for oil leaks. If oil leaks are evident, replace the auto-tensioner.

(3) Insert a 1.4 mm (0.055 in.) wire in the set hole





② Align the timing mark of the intake camshaft sprocket and place the timing belt around that sprocket. Then, clip the belt at the location shown.

Caution

- **The camshaft sprockets can turn easily and do not give excessive tension to the timing belt.**

③ Place the timing belt around the idler pulley.

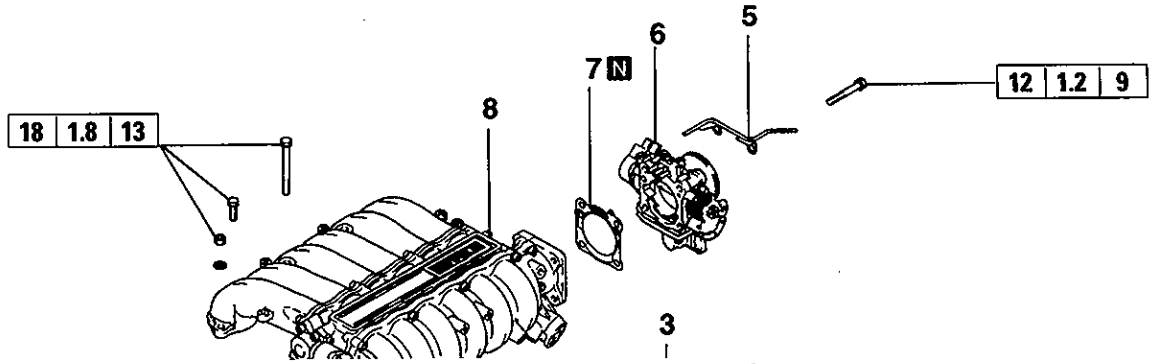
④ On the left bank, make sure that the timing marks of the camshaft sprockets are aligned and then hold the timing belt on these sprockets with paper clips.

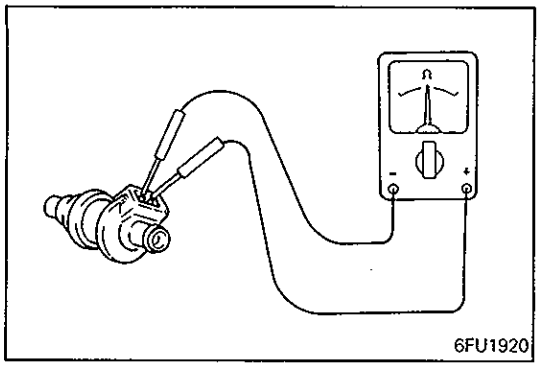
⑤ Place the timing belt around the water pump pulley.

NOTES

5. FUEL AND EMISSION CONTROL PARTS

REMOVAL AND INSTALLATION





INSPECTION

INJECTOR

(1) Using a circuit tester, measure the resistance of each injector.

Standard value: 13 – 16 Ω

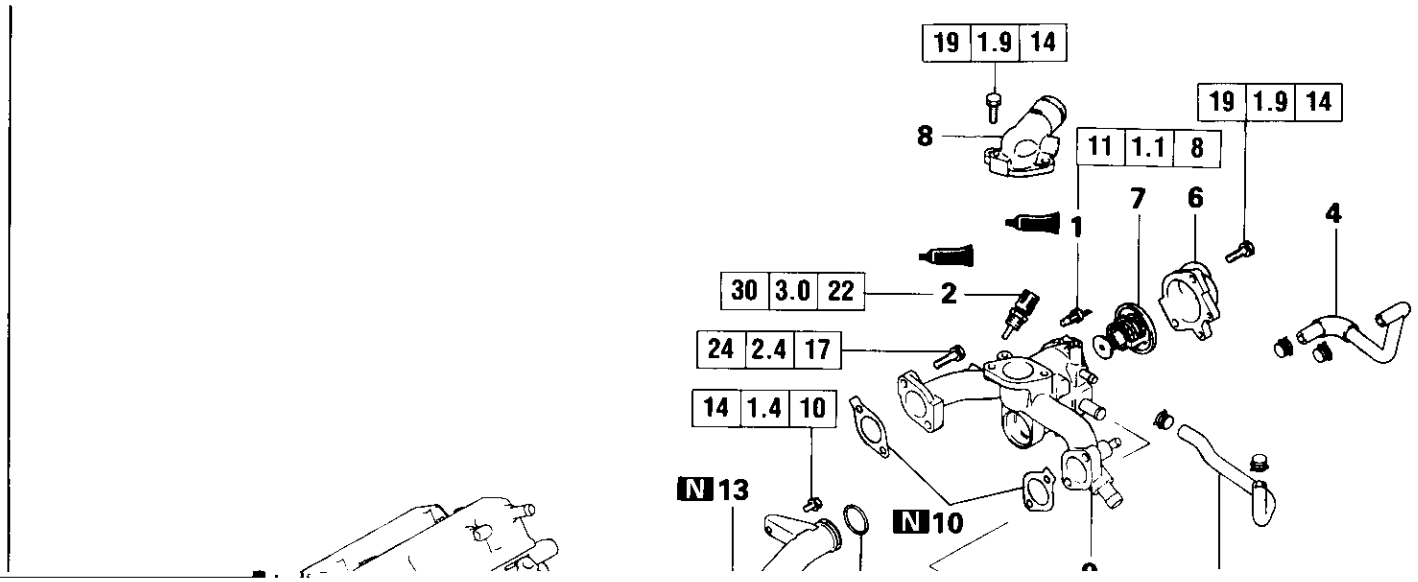
6. IGNITION SYSTEM

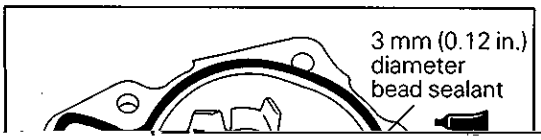
REMOVAL AND INSTALLATION

NOTES

7. WATER PUMP AND WATER PIPE

REMOVAL AND INSTALLATION

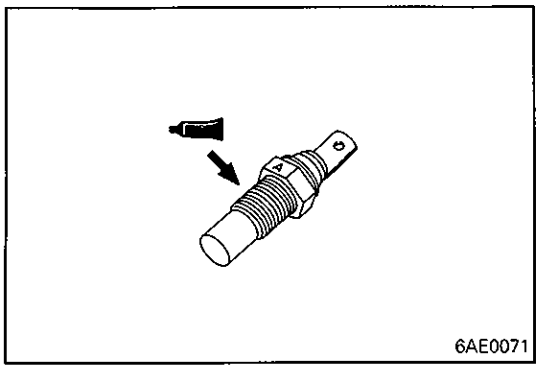




INSTALLATION SERVICE POINTS

▶A◀ SEALANT APPLICATION TO WATER PUMP

Specified sealant:

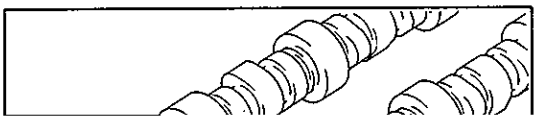


◆F◆ SEALANT APPLICATION TO ENGINE COOLANT TEMPERATURE GAUGE UNIT

Specified sealant:
3M ATD Part No. 8660 or equivalent

© Mitsubishi Motors Corporation Dec. 1992

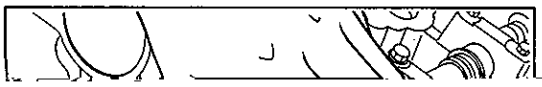
NOTES



REMOVAL SERVICE POINTS

◆A◆ CAMSHAFT INSTALLATION

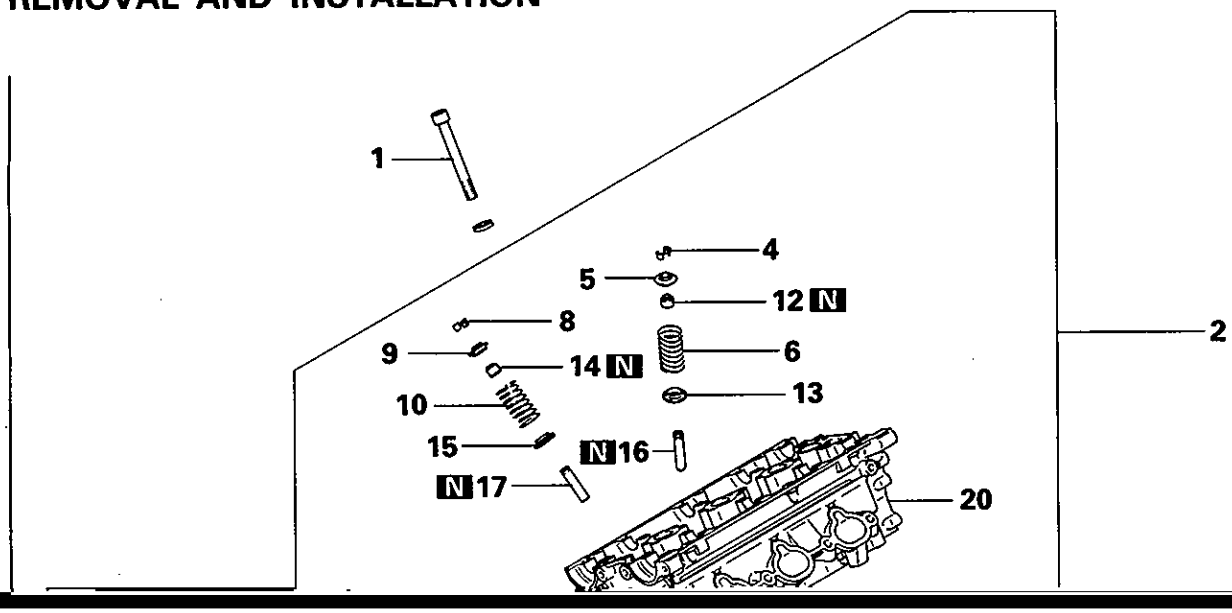
(1) Apply engine oil to journals and cams of the camshaft



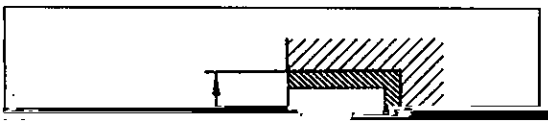
◆E◆ CIRCULAR PACKING INSTALLATION

10. CYLINDER HEAD AND VALVES

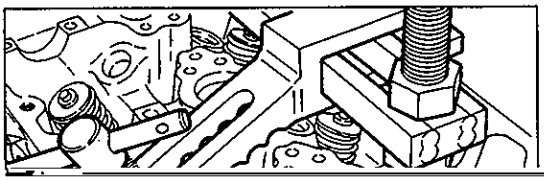
REMOVAL AND INSTALLATION



VALVE SPRING



(3) Before fitting the valve seat, either heat the cylinder head up to approximately 250°C (482°F) or cool the valve seat in liquid nitrogen, to prevent the cylinder head bore from galling.



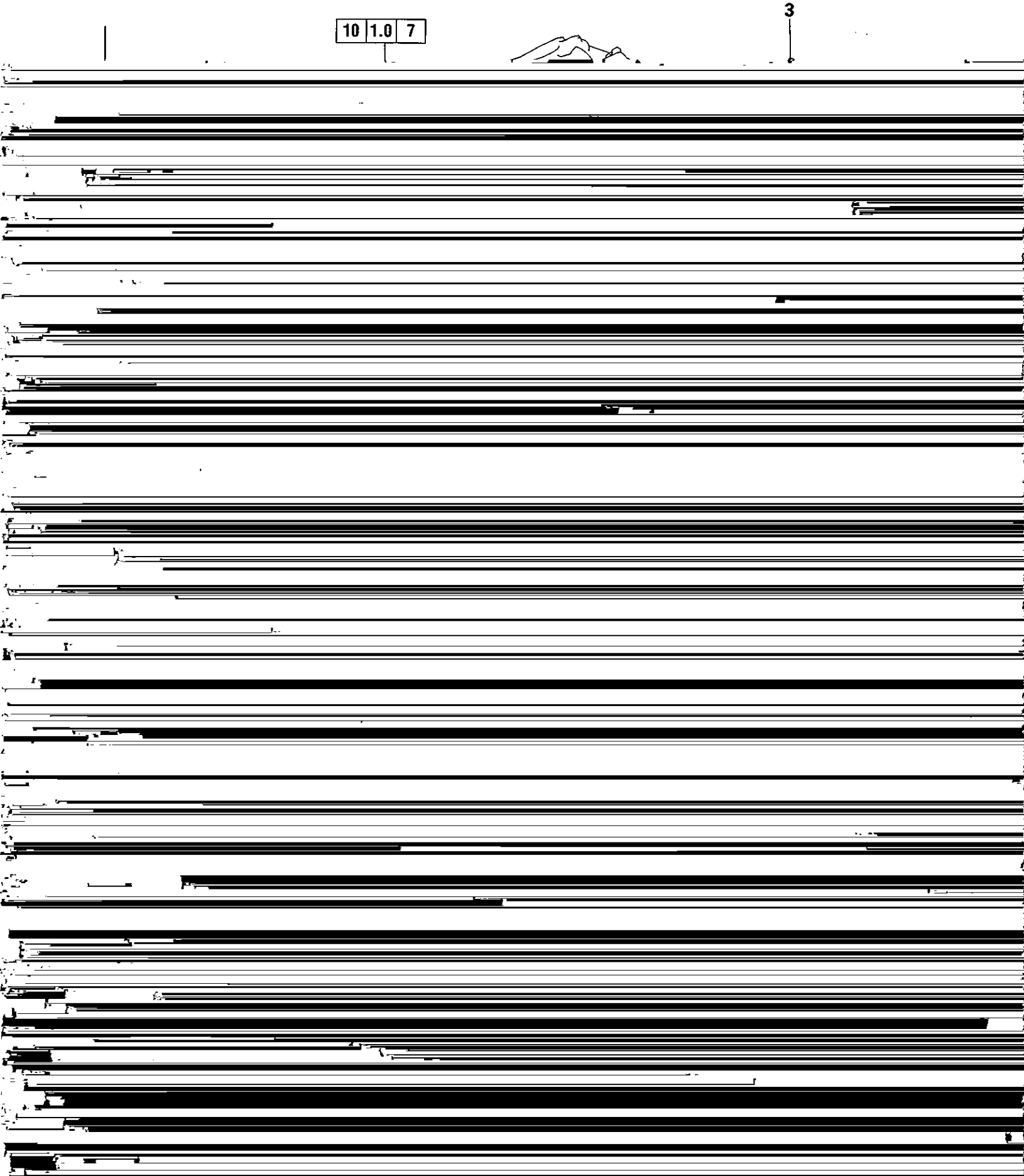
◆C◆ RETAINER LOCK INSTALLATION

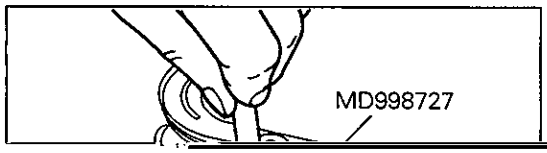
(1) The valve spring, if excessively compressed, causes the bottom end of retainer to be in contact with, and damage, the stem seal.

NOTES

11. OIL PUMP CASE AND OIL PAN

REMOVAL AND INSTALLATION

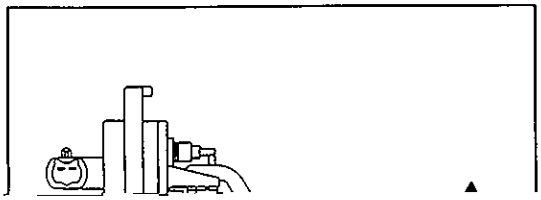




REMOVAL SERVICE POINTS

⟨A⟩ OIL PAN REMOVAL

(1) Knock the special tool deeply between the oil pan and the cyl-

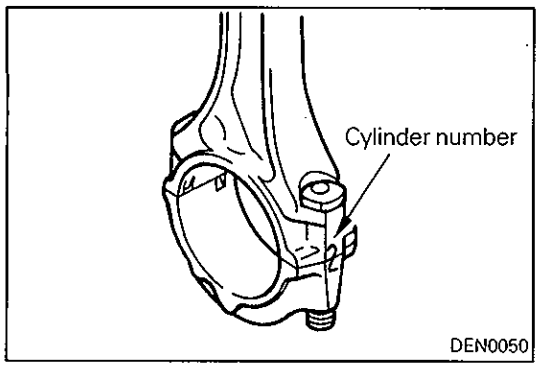


OIL LEVEL SENSOR (FOR EUROPE, GCC)

(1) Put the oil level sensor in the oil, then move the float up and down with the oil at a temperature either lower than 40°C (104°F) or higher than 80°C (176°F), and check for continuity.

12. PISTON AND CONNECTING ROD

REMOVAL AND INSTALLATION



REMOVAL SERVICE POINTS

◇A◇ CONNECTING ROD CAP REMOVAL

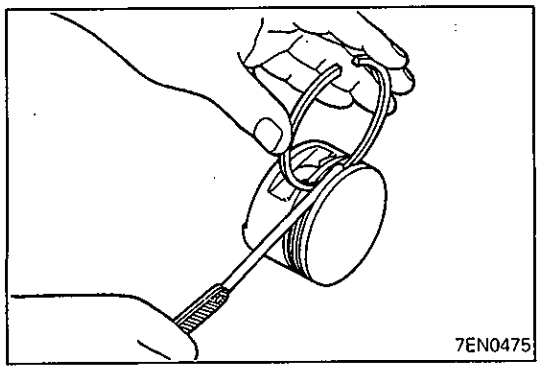
(1) Mark the large end of the connecting rod with the cylinder number for use during reassembly.

Piston pin setting tool MD998780

Push rod

Guide A

◇B◇ PISTON PIN REMOVAL



INSPECTION

PISTON RING

(1) Check the side clearance between the piston ring and ring groove. If the limit is exceeded, replace the ring or piston, or both.

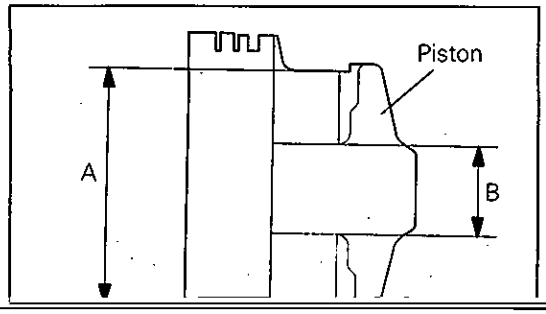
Standard value:

No. 1	**0.03 – 0.07 mm (0.0012 – 0.0028 in.)**
No. 2	**0.02 – 0.06 mm (0.0008 – 0.0024 in.)**

Limit:

No. 1	**0.1 mm (0.004 in.)**
No. 2	**0.1 mm (0.004 in.)**

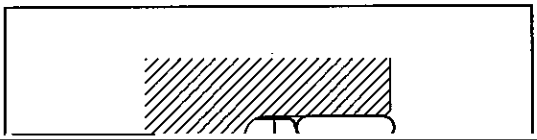
[illegible] piston ring into the cylinder bore. Force it down



INSTALLATION SERVICE POINTS

◆A◆ PISTON PIN INSTALLATION

(1) Measure the following dimensions of the piston, piston pin and connecting rod.
A: Piston pin insertion hole length
B: Distance between piston bosses
C: Piston pin length
D: Connecting rod small end width



◆B◆ OIL RING INSTALLATION

(1) Fit the oil ring spacer into the piston ring groove.

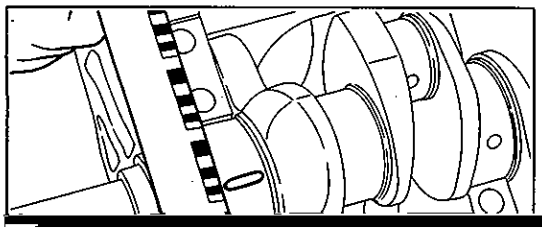
NOTE

The side rails and spacer may be installed in either direction.

(4) Use suitable thread protectors on connecting rod bolts before inserting piston and connecting rod assembly into cylinder block. Care must be taken not to nick crank pin.
(5) Using a suitable piston ring compressor tool, install piston

(2) Check if the thrust clearance in the connecting rod big end

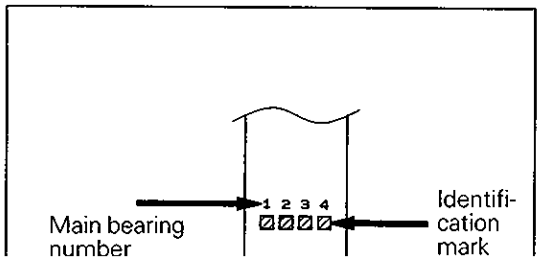
NOTES



INSPECTION

CRANKSHAFT OIL CLEARANCE (PLASTIC GAUGE METHOD)

(1) Remove oil from crankshaft journal and crankshaft bearing.
(2) Install the crankshaft.
(3) Cut the plastic gauge to the same length as the width of

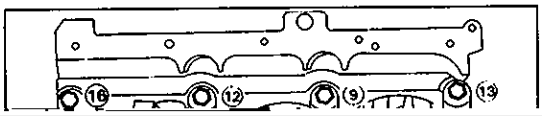


INSTALLATION SERVICE POINTS

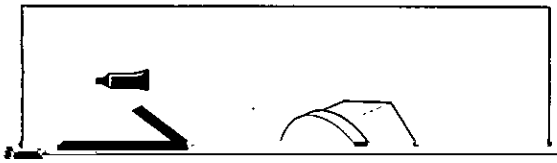
◆A◆ CRANKSHAFT BEARING INSTALLATION

(1) When replacing bearing, select a one of proper size according to the identification marks stamped on both crankshaft and cylinder block.

Crankshaft journal	Cylinder block bearing bore	Crankshaft bearing



(3) Apply engine oil to the threaded portion and bearing surface of the bolt.
(4) Tighten the bearing cap bolts to 25 Nm (2.5 kgm, 18 ft.lbs.)



◆E◆ DRIVE PLATE BOLT / FLYWHEEL BOLT INSTALLATION

(1) Remove all the remaining sealant from bolts and thread holes of crankshaft.

Service Bulletins

Click on the applicable bookmark to select the Service Bulletin.

SERVICE BULLETIN

QUALITY INFORMATION ANALYSIS
OVERSEAS SERVICE DEPT. MITSUBISHI MOTORS CORPORATION

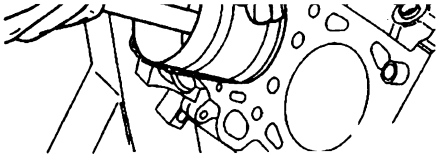
SERVICE BULLETIN	**No.:** MSB-00E11-003		
	Date: 2001-04-15	**<Model>**	**<M/Y>**
Subject:	CHANGE TO DESCRIPTIONS OF CONNECTING ROD BEARING SELECTION	(EC)GA-LANT(E50-80)	95-96
Group:	ENGINE	**Draft No.:** 00EN612016	
INFORMATION	**INTERNATIONAL CAR ADMINISTRATION OFFICE**	[redacted] T.MASAKI-MANAGER TECHNICAL SERVICE PLANNING	

1. Description:

In the following Workshop Manual, a change has been made to descriptions of the procedure for selecting the 6A1 engine connecting rod bearings due to change in identification marks and stamping positions of the connecting rod bearings.

2. Applicable Manuals:

Manual	Pub. No.	Language	Page(s)



(4) Use suitable thread protectors on connecting rod bolts before inserting piston and connecting rod assembly into cylinder block. Care must be taken not to nick crank pin.

(5) Using a suitable piston ring compressor tool, install piston and connecting rod assembly into the cylinder block.

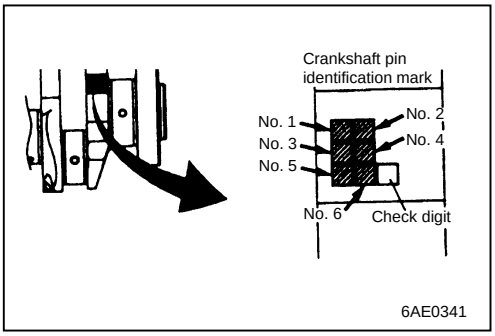
Caution

- **Insert the front mark (arrow) on the top of the piston so it faces the engine front (timing belt side)**

CONNECTING ROD BEARINGS INSTALLATION

(1) When the bearings are to be replaced, select correct ones and install them in the correct positions according to the identification marks stamped on the crankshaft.

Crankshaft		Connecting rod bearing	
Pin O.D. identification mark	Pin O.D. mm (in.)	Identification mark	Thickness mm (in.)
1	42.995-43.000		



Type 2

Crankshaft		Connecting rod bearing	
Pin O.D. identification mark	Pin O.D mm (in.)	Identification mark	Thickness mm (in.)
I	42.995-43.000 (1.6927-1.6929)	1	1.487-1.491 (0.0585-0.0587)
II	42.985-42.995 (1.6923-1.6927)	2	1.491-1.495 (0.0587-0.0589)
III	42.980-42.985 (1.6921-1.6923)	3	1.495-1.499 (0.0589-0.0590)